CASE STUDY 5: TEAM WORK-SPLIT & EXECUTION CHARTER

Scenario: POP3 Mailbox Access in a Low-Bandwidth Branch • **Framework:** 5-Member Equitable Task Distribution

Notice to All Team Members: This document provides the official task delegation, individual deliverables, and grading rubric alignment for our 5-member team on Case Study 5. A complete, high-quality master report (*'Case Study 5 - POP3 Mailbox Access in a Low-Bandwidth Branch.docx'*) has already been drafted. Use this charter to understand your assigned ownership, master your section for faculty viva/cross-questioning, and guarantee top marks (100/100).

1. Executive Team Responsibility Matrix

Direct mapping of each member to report sections, key deliverables, and viva specialization:

Member	Designated Role	Sections Owned	Core Deliverables & Tasks	Viva Focus
Member 1 (Team Lead)	**Lead Protocol Analyst & FSM Architect**	• Section 1 (Overview) • Section 6.1 (Question a)	• RFC 1939 Protocol Architecture & FSM states • Deconstruct POP3 limitations (monolithic transfer, lack of resume, flag loss) • Lead coordination & master doc merge	RFC 1939 FSM, POP3 3-phase lifecycle, broken RETR mechanics
Member 2	**WAN Systems & Workflow Architect**	• Section 1.1–1.4 • Section 6.2 (Question b)	• WAN link model (256 kbps, 450ms RTT, satellite) • 5-Tier Resilient Email Access Workflow • Store-and-forward edge proxy & TOP pre-fetch • Workflow ASCII sequence diagram	Why raw IMAP fails on VSAT, 5-Tier edge workflow, header pre-fetching
Member 3	**Simulation & Packet Capture Specialist**	• Section 5 (All) • Table 4 (Setup) • Subsections 5.1–5.3	• Wireshark trace breakdown (Port 110 vs 995) • Packet-level proof of -ERR [IN-USE] lockout • Python socket emulation script verification • Network topology diagram	Wireshark trace frames, TCP 3-way handshake, keepalive lockouts
Member 4	**Cybersecurity & Threat Analyst**	• Section 6.3 (Question c) • Threat Modeling • Hardening Plan	• Threat analysis of legacy mail (sniffing, MITM) • Cryptographic failure of APOP (MD5 collisions) • Hardening blueprint: TLS 1.3, SASL OAuth 2.0 • Reliability failure modes (lock freezes, PST loss)	Cleartext port 110 sniffing, APOP MD5 weakness, OAuth2/TLS 1.3
Member 5	**Quality, Benchmarks & Standards Lead**	• Sections 2, 3 (Tables) • Section 4 (Findings) • Section 5.4, 7 & 8	• Populate Table 2 (Protocols) & Table 3 (RFCs) • Section 5.4 Comparative Performance Table • Base64 37% MIME expansion calculations • Section 7 Conclusion & IEEE Reference audit	Base64 37% penalty, POP3 vs IMAP benchmarks, RFC citations

2. Individual Role Profiles & Actionable Tasks

Each member must read and fully understand their designated technical area:

MEMBER 1: LEAD PROTOCOL ANALYST & FSM ARCHITECT

Primary Ownership: Section 1 (Case Study Overview: Background, Problem Statement, Objectives, Scope, Key Concepts) & Section 6.1 (Question a: Explain POP3 and Its Limitations).

Specific Actionable Tasks:

- **Deconstruct RFC 1939 Architecture:** Explain client-server text-based dialogue over TCP port 110/995, command verbs (STAT, LIST, RETR, DELE, TOP, UIDL, QUIT), and +OK / -ERR reply tokens.
- **Map the 3-State Finite State Machine:** Detail AUTHORIZATION (USER/PASS/APOP), TRANSACTION (mailbox interrogation), and UPDATE (commit deletions upon QUIT).
- **Analyze Core Protocol Limitations:** Explain why monolithic retrieval chokes narrow pipes, why RETR cannot resume broken byte offsets, and why abrupt disconnects trigger non-atomic deletion rollbacks (causing duplicate emails).
- **Team Leadership:** Ensure all team members' details are entered into Table 1 of the Word document, conduct final integration, and monitor milestone deadlines.

MEMBER 2: WAN SYSTEMS & WORKFLOW ARCHITECT

Primary Ownership: Section 1.1–1.4 (WAN Impairment Modeling) & Section 6.2 (Question b: Design an Appropriate Email-Access Workflow).

Specific Actionable Tasks:

- **Model Constrained WAN Environment:** Quantify transmission latency over geostationary satellite (VSAT) links (450–700 ms RTT, 256 kbps throughput, 5% random packet loss, rain fade).
- **Design the 5-Tier Resilient Email Architecture:**

- *Tier 1 (LAN Isolation):* Deploy local branch edge mail gateway (Dovecot/Fetchmail proxy) providing instant LAN speeds.
- *Tier 2 (Header Pre-Fetch):* Use TOP msg 0 to triage subject, sender, and size without pulling payloads (<15 KB sync).
- *Tier 3 (Attachment Throttling):* Auto-sync <100 KB text; hold >100 KB MIME attachments at HQ until on-demand click.
- *Tier 4 (Wire Compression):* Implement zlib/gzip tunnel compression, slashing text/MIME payloads by 65%.
- *Tier 5 (Outbound Spooling):* Local Postfix relay queuing outbound mail with exponential backoff retry timers.

- **Illustrate Workflow Sequence:** Walk through the ASCII flowchart showing Client MUA <--> Local Proxy <--> WAN <--> HQ Mail Server.

MEMBER 3: SIMULATION & PACKET CAPTURE SPECIALIST

Primary Ownership: Section 5 (Implementation / Simulation / Demonstration), Table 4, Subsections 5.1 (Topology), 5.2 (Wireshark Trace), and 5.3 (Python Socket Verification).

Specific Actionable Tasks:

- **Configure Simulation Parameters (Table 4):** Document tools (Wireshark v4.2.x, Python 3.12 socket/poplib, Ubuntu Linux tc-netem) and experimental setup (192.168.10.50 MUA <--> 256 kbps WAN <--> 10.0.0.25 HQ server).
- **Analyze Wireshark Protocol Traces:** Break down frame-by-frame interactions: TCP 3-way handshake (452ms RTT), cleartext PASS exposure in Frame 07, streaming RETR 4, simulated link severance, and TCP RTO backoff.
- **Demonstrate Mailbox Lock Contention:** Highlight Frame 94 showing reconnection failure: -ERR [IN-USE] Mailbox is locked by another session due to orphaned half-open TCP sockets.
- **Verify Python Socket Harness:** Explain how the Python socket script successfully extracts headers via TOP 4 0 in 820 bytes, saving 1.88 MB of WAN transfer.

MEMBER 4: CYBERSECURITY & THREAT ANALYST

Primary Ownership: Section 6.3 (Question c: Analyze Security and Reliability Concerns of Legacy Mail Access) & Hardening Blueprint.

Specific Actionable Tasks:

- **Audit Security Vulnerabilities:** Detail cleartext sniffing on TCP port 110, credential theft, man-in-the-middle (MITM) tampering, and lack of message integrity verification.
- **Cryptographic Breakdown of APOP:** Explain why RFC 1939 APOP is obsolete—MD5 hash collisions, length-extension vulnerabilities, and susceptibility to offline rainbow tables.
- **Identify Operational Failure Modes:** Explain exclusive lock freezing (half-open TCP sessions keeping dotlock/flock engaged for 15–30 mins), duplicate delivery storms, and local PST/mbox file corruption risks.
- **Formulate Enterprise Hardening Plan:** Detail mandatory migration to POP3S (Port 995 over TLS 1.3 with TLS_AES_256_GCM_SHA384), SASL XOAUTH2 token-based authentication (MFA integration), and DMZ perimeter firewall access lists.

MEMBER 5: QUALITY, BENCHMARKS & STANDARDS LEAD

Primary Ownership: Section 2 (Technologies Table), Section 3 (Standards/RFC Table), Section 4 (Research Findings), Section 5.4 (Benchmark Matrix), Section 7 (Conclusion), and Section 8 (References).

Specific Actionable Tasks:

- **Complete Table 2 & Table 3:** Rigorously define POP3, IMAP4, SMTP, POP3S, and MIME; verify RFC citations (RFC 1939, RFC 2595, RFC 3501, RFC 5321, RFC 8446, RFC 7628) with valid URLs and access dates.
- **Calculate MIME Bandwidth Penalties:** Provide mathematical proof of Base64 encoding overhead (3 binary bytes $\rightarrow$ 4 ASCII characters = 33.33% inflation, ~37% with CRLF wrapping). Show why a 3 MB attachment inflates to 4.11 MB.
- **Evaluate Performance Benchmarks (Table 5):** Contrast Legacy POP3 (3.15 MB / 118s) vs IMAP4 (480 KB / 42s) vs Proposed Edge Workflow (14 KB / 1.9s) across bandwidth, completion time, and recovery.
- **Synthesize Conclusion & References:** Detail the 3-phase strategic roadmap (Immediate proxy $\rightarrow$ Offline IMAP $\rightarrow$ SD-WAN with FEC) and ensure IEEE citation formatting compliance.

3. Step-by-Step Procedure & Execution Timeline

Follow this structured 4-phase timeline to guarantee seamless collaboration without overlapping work:

Phase & Timeline	Core Objectives	Actionable Steps & Milestone Checks	Responsible
PHASE 1 Day 1 (Kick-off)	Task Alignment & Background Research	• Hold 20-min team briefing to confirm role assignments. • Everyone reviews their assigned RFCs & syllabus concepts. • Open the master DOCX report and read assigned sections.	All 5 Members (Lead by M1)
PHASE 2 Day 2 (Evidence)	Simulation & Data Verification	• M3 reviews Wireshark traces, packet timings & socket logs. • M2 & M5 verify bandwidth calculations (37% Base64 inflation). • M4 finalizes security vulnerability matrix & cipher suites.	M2, M3, M4, M5
PHASE 3 Day 3 (Integration)	Master Document Personalization	• Enter all 5 team member names, registration numbers & roles in Table 1. • Enter Course Instructor name and Team Number on cover. • Ensure all sub-bullet points and diagrams render cleanly.	M1 (Lead) & M5 (Review)
PHASE 4 Day 4 (Defense)	Peer Review & Viva Preparation	• Conduct internal mock viva using Section 4 questions. • Verify full compliance with 100-mark evaluation rubric. • Export final PDF alongside DOCX for LMS submission.	All 5 Members

4. Faculty Viva & Defense Cheat-Sheet (High-Probability Questions)

Every member must memorize the bulletproof answer to their question for cross-examination:

[Member 1 Question] Q: ***Why can't standard POP3 resume an interrupted download if the link drops at 90%?***

A (30-Second Bulletproof Defense): POP3 (RFC 1939) specifies the RETR command without byte-offset parameters, unlike HTTP Range headers. Furthermore, message state commits exclusively during the UPDATE phase upon a clean QUIT. A sudden TCP drop rolls back uncommitted flags, forcing the client to re-request the entire payload from byte zero.

[Member 2 Question] Q: ***Why not simply switch the branch to IMAP4 instead of building a store-and-forward proxy?***

A (30-Second Bulletproof Defense): Standard IMAP4 is a highly chatty protocol requiring multiple bidirectional round-trips for mailbox state and flag synchronization. Over a geostationary satellite link with 450–700 ms RTT, this causes severe UI lag and consumes continuous airtime. Our edge store-and-forward proxy combines offline local caching with selective header-only WAN fetching via TOP, cutting WAN sync traffic by over 98%.

[Member 3 Question] Q: ***What packet-level evidence proves that POP3 causes denial of service after link disruption?***

A (30-Second Bulletproof Defense): In Frame 86 of our Wireshark capture, an ungraceful link disconnect leaves a half-open TCP socket without sending FIN/RST. The server daemon retains an exclusive file lock (dotlock/flock). In Frame 94, when the user reconnects, the server rejects authentication with '-ERR [IN-USE] Mailbox is locked' until the TCP keepalive timer expires (15–30 mins).

[Member 4 Question] Q: ***Why is APOP considered cryptographically obsolete despite not sending plaintext passwords?***

A (30-Second Bulletproof Defense): APOP relies on an MD5 challenge-response digest: MD5(timestamp + shared_secret). MD5 is cryptographically compromised due to practical collision and length-extension attacks. Moreover, APOP cannot authenticate the server, lacks PFS (Perfect Forward Secrecy), and cannot support modern Multi-Factor Authentication or OAuth 2.0 tokens.

[Member 5 Question] Q: ***How does MIME Base64 encoding penalize bandwidth in a 256 kbps branch network?***

A (30-Second Bulletproof Defense): Base64 converts 3 binary octets (24 bits) into 4 ASCII characters (32 bits), creating an inherent 33.33% data inflation. When RFC 2045 line wrapping (CRLF every 76 characters) is included, expansion reaches ~37%. A 3 MB PDF attachment expands to 4.11 MB, monopolizing a 256 kbps link for 147 seconds under ideal conditions.

5. Evaluation Rubric & Pre-Submission Checklist

Assessment Rubric Criteria	Marks	How Our Team Secures Maximum Marks	Lead Owner
Conceptual Understanding & Application	**20**	Deep explanation of POP3 FSM, client-server mail architecture, and TCP transport behavior over degraded links.	Member 1
Quality & Relevance of Research	**15**	Direct citations of RFC 1939, 2595, 3501, 5321, 8446 with exact URLs, access dates, and vendor docs (Dovecot/Postfix).	Member 5
Security & Technical Analysis	**20**	Exhaustive threat model of Port 110 sniffing, APOP MD5 collision attacks, lock freezes, and TLS 1.3 / OAuth2 hardening.	Member 4
Architecture, Design & Packet Evidence	**20**	5-tier edge proxy workflow design + frame-by-frame Wireshark packet capture trace + Python socket emulation log.	Member 2 & 3
Justification and Conclusions	**15**	Quantitative benchmark table (Table 5) proving 98% bandwidth reduction + 3-phase strategic modernization roadmap.	Member 5
Presentation, Diagrams & Formatting	**10**	Strict adherence to SCOPE DOCX template, Times New Roman typography, clear ASCII topologies, and zero placeholders.	Member 1 & 5
TOTAL EVALUATION SCORE	**100**	**Converted to 10 Marks in Final Academic Weightage**	**ALL TEAM**

FINAL SUBMISSION CHECKLIST (BEFORE UPLOADING TO LMS):

[ ] **Master File:** *'Case Study 5 - POP3 Mailbox Access in a Low-Bandwidth Branch.docx'* is ready in workspace.
[ ] **Table 1 (Team Details):** All 5 student registration numbers, names, and assigned roles are entered.
[ ] **Table 0 (Cover Block):** Course Instructor name and Team Number are completed.
[ ] **Readiness:** Every member has reviewed their assigned viva defense question above.
[ ] **File Export:** Both DOCX and PDF versions are exported for group sharing and submission.